

Michael John Heap

Position Full professor at the University of Strasbourg (France)

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Nationality English

Age 43

Main research theme: I am an experimental volcanologist who focuses on understanding the physical, mechanical, and transport properties of volcanic rocks, with an emphasis on understanding the influence of hydrothermal alteration on these properties and, as a result, volcanic hazards. My work typically combines experimental work with field work and modelling.

Other research themes: the mechanical behaviour and failure modes of porous rocks; permeability; time-dependent deformation; geothermal energy; planetary science; building/monument conservation and preservation; concrete and cement

Notable scientific contributions: My main scientific contribution is the fusion of experimental rock deformation and volcanology, which has led to significant advancements in our understanding of volcanic hazards such as flank collapse and explosive eruptions. More recently, and using a multidisciplinary approach, I have been exploring the link between hydrothermal alteration and volcanic hazards. This research has led to numerous scientific publications in prestigious journals, a National French Research Agency (ANR) grant, my induction into the Academic Institute of France (IUF), and a European Research Council (ERC) SYNERGY grant worth ~10 million Euros.

1. Employment and Education

2022-onwards Full professor at the School and Observatory of Earth Sciences (EOST) at the University of Strasbourg (France)

2010–2022 Associate professor at EOST, University of Strasbourg (France)

2021–2026 Elected as member of the Academic Institute of France (IUF)

2021 Invited professor (four months) at Swiss Federal Institute of Technology (EPFL, Switzerland)

2018 Habilitation to supervise research (HDR), University of Strasbourg (France)

2009–2010 Post-doc at Ludwig Maximilians University Munich (Germany)

2005–2009 Ph. D at University College London (UK)

2001–2005 First class MSci in geology at the University of Liverpool (UK)

2. Teaching experience, service, and outreach

- 16 years of teaching experience as associate professor/full professor
- Teaching outreach: I have taught courses in Switzerland, Azerbaijan, China, and New Zealand
- Given four workshops on paper writing: two in China, one at the Volcanic and Magmatic Studies Group (VSMG) annual meeting (2023), and one for the "IAVCEI ECR-Net writing masterclass" (2023)
- International relations correspondent (CORI) for EOST; responsible for international mobility and attractivity at EOST

- In charge of two courses at EOST: geophysical measurements in the laboratory 1A; geophysical measurements in the laboratory 2A
- 2020–2021: coordinator for the Master (M1 and M2) Geoscience programs at the French-Azerbaijani University (UFAZ); in charge of final year internships at UFAZ
- 2020–2021: in charge of three courses at UFAZ: hydrological modelling 2 (UFAZ M2); geoscience seminars (UFAZ M2); project work 2 (UFAZ M2)

3. Research supervision, service, and outreach

- I have supervised many masters research projects, engineering diploma projects, and undergraduate research projects at EOST
- Supervised/supervising 13 Ph. D students, 6 post-docs, 1 research engineer
- Examined 11 Ph. D theses; reporter for one assistant professor position; examined one HDR (habilitation à diriger des recherches) thesis
- Editor-in-Chief for Journal of Volcanology and Geothermal Research (2025–)
- Associate Editor for Journal of Geophysical Research: Solid Earth (2018–2025). I made decisions >100 manuscripts
- Editorial Board for Journal of Volcanology and Geothermal Research (2018–2025); Editor for, and founding member of, diamond-open-access journal Volcanica (2018–2024); Topical Editor for Solid Earth (2012–2020)
- Organised and chaired 13 sessions at the European Geosciences Union meeting (Vienna, Austria); currently main convenor for long-running session "Volcanic processes: tectonics, deformation, geodesy, unrest"
- Member of the American Geophysical Union, the European Geosciences Union, and the International Association of Volcanology and Chemistry of the Earth's Interior
- Scientific coordinator/PI for several large grants
- I have reviewed large national and international grants

4. Honours: grants, awards, and medals

- Awarded >11.5 M Euros in research grants (as PI)
- Won **European Research Council (ERC) SYNERGY grant** (2024–2030) worth 9989653 Euros to study the role of hydrothermal alteration on volcano morphology, instability, and unpredictable volcanic hazards
- Won **National French Research Agency (ANR) grant** (2022–2026) worth 673344 Euros to study the role of hydrothermal alteration on the stability of La Soufrière de Guadeloupe (Eastern Caribbean)
- Won the **2025 VMSG Award** of the Volcanic and Magmatic Studies Group, a joint special interest group of the Mineralogical Society of Great Britain and Ireland & The Geological Society of London. Citation: "Mike has been an active member of the VMSG community for many years and amongst his substantial and interdisciplinary academic contributions to volcanology he has also been a strong advocate for open science, equitable collaboration and in supporting early career researchers globally – all of which are strongly in the spirit of the award"
- Won the **2023 Wager Medal** of the International Association of Volcanology and Chemistry of the Earth's Interior (IAVCEI) for "outstanding contributions to volcanology"
- Awarded travel grants to work with colleagues in Ireland, UK, Switzerland, Germany, Italy, China, and New Zealand
- Most-cited Bulletin of Volcanology paper of 2021 Award (based on 2020 and 2021 citations) for the paper "The mechanical behaviour and failure modes of volcanic rocks: a review"

- **Academic Institute of France (IUF) laureate** (Junior Member) in 2021–2026, a service of the French Ministry of Higher Education that “distinguishes each year a small number of university professors for their research excellence, as evidenced by their international recognition”
- Written three review papers (two invited): *Journal of Structural Geology* (2013), *Journal of Volcanology and Geothermal Research* (2020), and *Bulletin of Volcanology* (2021)
- **Initiative d'Excellence Prix Scientifiques 2017**: “Les Espoirs de l'Université de Strasbourg”. This medal is to “reward scientists who have shown originality and dynamism in the implementation of their research and whose future developments look promising”
- Most-cited *Bulletin of Volcanology* paper of 2017 Award (based on 2015 and 2016 citations) for the paper “Fracture and compaction of andesite in a volcanic edifice”
- Awarded the “Prime d'Excellence Scientifique (PES)” at the Université de Strasbourg in 2012, 2017, and 2021 (an initiative designed to reward teaching staff with a high research output)
- Solicited/keynote speaker at the American Geophysical Union (twice), European Geosciences Union (three times), the Deformation, Rheology, and Tectonics (DRT) meeting, China Rock (twice), and the International Conference on Earth Energy Science

5. Research output

- *h*-index 68
- 244 papers in peer-reviewed journals
- >14000 citations
- 1 paper cited >800 times; 2 papers cited >500 times; 3 papers cited >400 times; 4 papers cited >300 times; 15 papers cited >200 times; 43 papers cited >100 times
- Published with 435 different co-authors

Selected recent publications:

Heap, M. J., & Violay, M. E. (2021). The mechanical behaviour and failure modes of volcanic rocks: a review. *Bulletin of Volcanology*, 83(5), 33. [211 citations; awarded most-cited paper in BV 2021]

Heap, M. J., Baumann, T., Gilg, H. A., Kolzenburg, S., Ryan, A. G., Villeneuve, M., ... & Clynne, M. A. (2021). Hydrothermal alteration can result in pore pressurization and volcano instability. *Geology*, 49(11), 1348-1352. [102 citations]

Heap, M. J., Baumann, T. S., Rosas-Carbajal, M., Komorowski, J. C., Gilg, H. A., Villeneuve, M., ... & Reuschlé, T. (2021). Alteration-induced volcano instability at la Soufrière de Guadeloupe (Eastern Caribbean). *Journal of Geophysical Research: Solid Earth*, 126(8), e2021JB022514. [89 citations]

Heap, M. J., Troll, V. R., Kushnir, A. R., Gilg, H. A., Collinson, A. S., Deegan, F. M., ... & Walter, T. R. (2019). Hydrothermal alteration of andesitic lava domes can lead to explosive volcanic behaviour. *Nature Communications*, 10(1), 5063. [143 citations]

Heap, M. J., Farquharson, J. I., Baud, P., Lavallée, Y., & Reuschlé, T. (2015). Fracture and compaction of andesite in a volcanic edifice. *Bulletin of Volcanology*, 77, 1-19. [145 citations; awarded most-cited paper in BV 2017]